TETANUS IMMUNOGLOBULIN

Immediate passive protection

Different site from vaccine

CHEMICAL NAME Human tetanus immunoglobulin (HTIG) — IgG antibodies against tetanus toxin from pooled plasma of immunised donors

CLASS Specific human immunoglobulin (passive immunisation)

TRADE NAMES HyperTET S/D · Tetagam P · Igantet · Tetabulin

THERAPEUTIC USES / INDICATIONS

- Tetanus-prone wounds in patients with incomplete, unknown or no tetanus vaccination (high-risk wounds).
- Treatment of clinical tetanus.
- Immunocompromised patients with tetanus-prone wounds regardless of vaccination status.

ADMINISTRATION

- **IM** (deep) at a different site from tetanus vaccine; some products IV for treatment.
- Store at 2–8 °C.
- Blood product — record batch number.

DOSAGES

- **Prophylaxis:** 250 IU IM (500 IU if >24 h since injury, risk of heavy contamination, or burns).
- **Treatment of tetanus:** IV immunoglobulin 5,000 IU (<50 kg) or 10,000 IU (>50 kg) (UK: IVIG, as IM volume is too large).
- Same dose for children.

PHARMACOKINETICS

- **Onset:** immediate protective antibody levels. **Duration:** ~3–4 weeks.
- **t½:** ~3–4 weeks.

PHARMACODYNAMICS

- Provides ready-made **antitoxin antibodies** that neutralise circulating (unbound) tetanus toxin — cannot reverse toxin already bound to nerves.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain, tenderness.
- Low-grade fever.

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity/anaphylaxis (rare); IgA-deficient patients with anti-IgA antibodies.
- Theoretical infection transmission (plasma-derived, virus inactivated).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to human immunoglobulins.
- **Caution:** IgA deficiency, thrombocytopenia (IM).

KEY INTERACTIONS

- **Live vaccines (MMR, varicella):** reduced response; delay ≥3 months after immunoglobulin.
- **Tetanus vaccine:** give at a separate site (no interference).

NURSING MANAGEMENT

- Assess wound and immunisation history; give vaccine and HTIG at different sites.
- Wound debridement and cleaning.
- Clinical tetanus: ICU care — quiet dark room, airway, muscle spasm control (benzodiazepines, magnesium), metronidazole.
- Record batch number.

PATIENT EDUCATION

- Complete the tetanus vaccine course after the injection.
- Report fever, rash or swelling at the site.

KEY POINTS

- Passive antitoxin; immediate protection.
- Treatment: IVIG 5,000–10,000 IU by weight.
- Prophylaxis 250 IU IM (500 IU high risk).
- Different site from vaccine; delay live vaccines.

SOURCES CHECKED **[1007]** Human Tetanus Immunoglobulin (UK SmPC) **[1008]** UKHSA Green Book, ch. 30

Full references p. 432–457